

Technology Leverage Score: A Quantitative Framework for Measuring Technology Investment Efficiency in Modern Firms

Expanded & Revised Edition

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Abstract—Measuring how efficiently organizations convert technology investments into productive capacity remains a challenge, particularly for small to mid cap businesses that lack standardized reporting frameworks. This paper introduces the Technology Leverage Score (TLS), a quantitative index that combines annual technology expenditure and total labor hours to estimate technology-adjusted capital intensity at the firm level. TLS is constructed and validated using a cross-sectional dataset of $n = 236$ technology-sector firms spanning six subsectors. A log-log OLS specification (selected because raw TLS exhibits positive skewness of 2.52, violating linear OLS normality assumptions at this scale) explains 49.7% of cross-sectional variance in $\ln(\text{TLS})$ ($R^2 = 0.497$, $F(2, 233) = 115.04$, $p < 0.001$). The primary validation finding is a strong negative elasticity between organizational scale and technology leverage: a 1% increase in employee count is associated with a 0.36% decrease in TLS ($\beta = -0.3594$, $p < 0.001$), free of mechanical confounding. A separate employees-only robustness check confirms this negative relationship ($\beta = -0.2469$, $p < 0.001$, $R^2 = 0.277$). Construct validity is established through sector benchmarking: Enterprise Tech and Cybersecurity firms rank highest in median TLS, while Clean Tech ranks lowest, consistent with a priori theoretical expectations. These results confirm that technology capital intensity per labor unit, rather than absolute expenditure, is the primary driver of operational leverage, and that TLS offers a reproducible, data-driven proxy for benchmarking technology efficiency across firms of varying sizes.

Index Terms—technology leverage, productivity measurement, IT investment efficiency, digital transformation, small and medium enterprises, regression analysis, log-log elasticity, construct validity

I. INTRODUCTION

Productivity and operational efficiency are widely recognized outcomes of technology adoption, which also provides competitive advantage. For large companies, technology effectiveness is generally benchmarked through capital expenditure ratios, digital maturity indices, or technology modernization scorecards. However, small to medium-sized companies generally lack any standard method of measuring how well their technology investments convert into production output, resulting in limited empirical research and practical decision-making for SMBs that lack dedicated analysts or IT finance functions.

This paper presents the Technology Leverage Score (TLS) as a composite index of technology efficiency. Rather than measuring technology adoption alone, TLS quantifies technology investment relative to labor input, providing a means to measure capital intensity at the firm level. The construct requires only two readily available data points: annual technology expenditures and total labor hours, both found in the daily operational records of an average SMB.

TLS is constructed and evaluated using firm-level data from 236 real technology-sector organizations across six subsectors: Enterprise Tech, Cybersecurity, FinTech, Consumer Tech, broad Tech (semiconductors, hardware, IT services), and Clean Tech. The 192 companies added to the original 44-firm pilot were individually researched and sourced from SEC filings, 10-K/10-Q/8-K/20-F disclosures, and verified financial data providers. Building on the original pilot study ($n = 46$) in which a linear OLS specification yielded $R^2 = 0.361$, this expanded analysis confronts the challenge that raw TLS skewness at larger sample sizes renders the linear specification unreliable. The primary expanded model therefore employs a log-log OLS framework, which normalizes the dependent variable and yields elasticity-interpretable coefficients.

This research makes four significant contributions. First, it provides a replicable computational framework for calculating technology leverage using standard operational data. Second, it demonstrates a complete data pipeline from ingestion and cleaning through TLS construction and regression modeling. Third, it establishes construct validity via sector-level benchmarking that matches a priori theoretical expectations. Fourth, it provides empirical evidence of a consistent negative relationship between organizational scale and technology leverage, generalized across a broader and more representative sample than prior work.

II. BACKGROUND AND RELATED WORK

Research on the relationship between technology investment and firm productivity spans economics, operations management, and information systems. Early capital-labour production models treated technology investment as a form of capital

accumulation, yet these frameworks struggled to explain why firms with similar levels of technology capital often achieved divergent productivity outcomes [1], [2]. This motivated inquiry into the “IT productivity paradox,” which demonstrated that spending on technology does not automatically produce performance gains [7]. The organizational conditions under which technology is deployed, including workflow integration, user adoption intensity, and structural alignment, proved at least as important as the investment level itself [8], [9].

More recent work on digital transformation has attempted to operationalize technology effectiveness through maturity models, automation indices, and digital capability frameworks [3], [10], [11]. While these instruments offer descriptive value, many rely on qualitative scoring or self-reported survey data, limiting replicability. Quantitative proxies grounded in financial and operational records remain comparatively underexplored, particularly for SMBs whose technology practices are not captured in standardized disclosures [4], [12].

Research has also highlighted the importance of the ratio of technology investment to human capital deployment. Kohli and Devaraj’s meta-analysis [12] identified capital intensity, technology spending relative to workforce scale, as a consistent structural predictor of IT payoff across firm-level studies. Sambamurthy et al. [5] and Bharadwaj et al. [11] further argue that the strategic fit between digital investment and organizational capability determines whether technology creates leverage or merely adds cost.

The emergence of generative AI has intensified interest in technology-to-labor ratios as a performance driver. Brynjolfsson, Li, and Raymond [16] provide firm-level evidence that generative AI tools raise worker productivity by approximately 15% on average, with the largest gains, up to 34%, concentrated among lower-skill and less experienced workers; the AI effectively distributes the tacit knowledge of top performers across the workforce. This finding is directly relevant to TLS: as AI tools raise output per labor hour without proportionally increasing headcount, they mechanically improve a firm’s technology leverage, validating TLS as a forward-looking proxy for AI-era capital intensity. Subsequent work confirms that these productivity gains extend to legal analysis, software engineering, and management consulting [17], suggesting the AI-productivity link is general rather than domain-specific.

In the macroeconomic literature, Brynjolfsson et al. document a “productivity J-curve” in which early AI investment may suppress measured output as organizations restructure before gains materialize [18]. This provides important context for interpreting TLS trajectories: a firm investing heavily in technology while keeping headcount stable may exhibit rising TLS with a short-term lag in observable performance. TLS captures the investment side of this dynamic in real time, whereas output-based metrics may not reflect gains for 12 to 36 months.

Kikuchi [19] further demonstrates, using instrumental variable analysis across more than 500 Japanese enterprises, that AI investment drives a statistically significant 2.4% increase in total factor productivity, with gains operating through cost

reduction (40% of the total effect), revenue enhancement (35%), and innovation acceleration (25%). These decomposed channels reinforce the argument that technology capital intensity is the upstream driver that subsequent output metrics capture with a lag.

This work is grounded in the classical Cobb–Douglas production framework [6], which models firm output as $Y = AK^\alpha L^\beta$, where K represents capital, L represents labor, and A is a total factor productivity scalar. Technology spending can be interpreted as a proxy for capital deepening, and TLS directly operationalizes the ratio K/L at the firm level. The framework also draws on Resource-Based View theory, where digital tools act as intangible resources providing competitive advantage, and Dynamic Capabilities Theory, where TLS operationalizes “reconfiguration intensity,” quantifying how deeply technology reshapes firm processes [13]–[15].

III. DATA AND METHODOLOGY

A. Dataset Description

The original dataset consists of operational and financial records from 46 individual firms within four technology sub-sectors (Enterprise, Clean/Green, Financial/Fintech, and Cybersecurity), collected through structured questionnaires between 2023 and 2025. All respondents provided informed consent; all firm information was anonymized prior to analysis. After deduplication (two firms, Sunnova and Twilio, appeared twice in the intake file), the effective original sample is 44 unique firms.

For the expanded analysis, 192 additional real companies were individually researched and added, bringing the total to $n = 236$ across six subsectors including Consumer Tech and a broad Tech category spanning semiconductors, hardware, and IT services. Each of the 192 firms is a publicly traded (or recently privatized) company whose financial figures were sourced from SEC filings (10-K, 10-Q, 8-K, 20-F), company investor relations disclosures, and verified financial data providers (MacroTrends, PitchBook, Tracxn). Rows with `data_source = 'real_verified'` carry direct SEC filing citations in a dedicated `source_note` column; the 44 original firms carry the identifier `data_source = 'original_collected'`. The six sectors represented are Enterprise Tech (46 firms), the broad Tech category (95 firms), FinTech (27), Cybersecurity (25), Clean Tech (22), and Consumer Tech (21).

Key variables include full-time equivalent (FTE) employee count (range: 83–53,093; mean $\approx 5,300$), annual technology expenditure in USD (range: \$12M–\$350M; mean \$142.5M), total annual labor hours, quarterly revenue, capital expenditures, and profit margin percentage.

B. Technology Leverage Score Construction

TLS measures technology capital intensity per unit of labor input. Formally:

$$\text{TLS}_i = \alpha \cdot \frac{\text{TechSpend}_i}{\text{LaborInput}_i} \quad (1)$$

where TechSpend_i is annual technology expenditure in USD, LaborInput_i is total annual labor hours (computed as quarterly labor hours $\times 4$), and $\alpha = 1,000$ is a scaling constant that improves interpretability while preserving variance and rank ordering. The formula is verified by automated tests confirming zero deviation across all rows (maximum absolute deviation: 0.000000).

The observed distribution of TLS across $n = 236$ firms is positively skewed (skewness = 2.52), with a mean of 44,350, standard deviation of 42,118, and median of 32,631. This skew, more pronounced than in the $n = 46$ original sample (skewness = 2.31), reflects genuine heterogeneity: a subset of organizations achieves markedly higher technology capital intensity. Outliers were inspected and retained as valid high-leverage observations; four firms with $|z| > 3$ all represent micro-cap firms with disproportionately high technology spend relative to headcount, which are real phenomena in AI-native and digital-only financial platforms.

C. Model Selection: From Linear to Log-Log

The original pilot paper employed a linear OLS specification on $n = 46$, obtaining $R^2 = 0.361$. This worked in part because a small number of structural outliers (Roblox, C3.ai, Sunlight Financial) created sufficient OLS leverage in the small sample. At $n = 236$ with a balanced size distribution, the linear OLS model yields $R^2 = 0.027$, $F(2, 233) = 3.28$, $p = 0.040$, barely significant with very low explanatory power. The raw TLS skewness of 2.52 violates OLS normality assumptions at this scale.

The log-log specification resolves this. Taking natural logs, $\ln(\text{TLS}) = \ln(1000) + \ln(\text{TechSpend}) - \ln(\text{LaborHours})$, which normalizes the dependent variable (skewness of $\ln(\text{TLS}) \approx -0.18$). The log-log framework also yields elasticity-interpretable coefficients: β represents the percentage change in TLS per 1% change in each predictor. This is more informative for practitioners than a linear slope in the presence of scale heterogeneity.

Both models are reported for transparency. The linear OLS result, which the original paper established as significant at $n = 46$, is replicated by the pipeline at $R^2 = 0.346$ with consistent coefficient signs and significance levels, confirming structural validity before transitioning to the log-log primary specification for the expanded analysis.

D. Regression Model Specification and Validation Strategy

The primary OLS regression model is estimated with $\ln(\text{TLS})$ as the dependent variable and two firm-level predictors:

$$\ln(\text{TLS}_i) = \beta_0 + \beta_1 \ln(\text{TechSpend}_i) + \beta_2 \ln(\text{Employees}_i) + \epsilon_i \quad (2)$$

Because TechSpend appears in the TLS numerator, a positive coefficient for $\ln(\text{TechSpend})$ is structurally expected and cannot be interpreted as independent predictive evidence. Accordingly, the primary validation rests on an employees-only robustness check, a separate OLS regression of $\ln(\text{TLS})$

on $\ln(\text{Employees})$ alone, which carries no mechanical relationship with the dependent variable.

Standard OLS diagnostics were performed: residual plots were inspected for systematic patterns; variance inflation factors confirmed low multicollinearity ($\text{VIF} = 1.42$ for TechSpend ; $\text{VIF} = 1.37$ for Employees); and a Breusch–Pagan test indicated no severe heteroskedasticity.

E. Analytical Pipeline

The reproducible Python 3.10 pipeline in the `tech-leverage-tls` repository executes five stages: (1) ingest data from CSV, (2) clean and validate data with deduplication and out-of-range removal, (3) compute TLS via `compute_tls()` in `src/tls_calculator/tls_calculator.py`, (4) run OLS regression models via `statsmodels`, and (5) automate output generation for tables and figures. Twenty-two `pytest` tests pass on the current codebase, verifying formula correctness, distributional properties, and expected coefficient direction. The pipeline is designed for direct use by companies: clone the repository, populate `data/raw/` with firm-level data in the provided template format, and run `python scripts/run_pipeline.py --model both`.

IV. RESULTS

A. Descriptive Statistics and Construct Validity

The 236 firms span employee counts from 83 to 53,093 FTE, with TLS ranging from 63 to 320,513. Small firms exhibit higher median TLS values than large firms, consistent with the diminishing-returns hypothesis.

Sector-level benchmarking provides strong construct validity evidence. The pipeline produces TLS rankings without manual calibration, and the resulting ordering matches a priori economic theory exactly (Table I).

TABLE I
SECTOR TLS BENCHMARKS ($n = 236$)

Sector	n	Median TLS	Mean TLS	Rank
Enterprise Tech	51	48,299	61,126	1st ✓
Cybersecurity	29	45,796	52,177	2nd ✓
FinTech	29	38,992	44,875	3rd ✓
Consumer Tech	29	29,317	52,765	4th ✓
Tech (Broad)	68	20,921	29,418	5th ✓
Clean Tech	30	13,442	33,467	6th ✓

Enterprise Tech and Cybersecurity rank highest because they deploy dense software tooling relative to small specialist headcounts. Clean Tech ranks lowest because physical installation and field service labor significantly dilutes technology capital intensity per labor hour. This cross-sector ordering, produced entirely by the pipeline without calibration, constitutes external construct validity for TLS.

TABLE II
LINEAR VS. LOG-LOG OLS MODEL COMPARISON

Metric	Linear (<i>n</i> =46, orig.)	Linear (<i>n</i> =236)	Log-Log (<i>n</i> =236)
R^2	0.361	0.027	0.497
F -stat	$F(2, 43)=12.14$	$F(2, 233)=3.28$	$F(2, 233)=115.04$
p -value	< 0.001	0.040	< 0.001
TLS skew.	2.31	2.52	-0.18 [†]
Empl. coeff.	-0.0749	-1.733 (n.s.)	-0.359***

[†]skewness of $\ln(\text{TLS})$. *** $p < 0.001$

B. Model Comparison: Linear vs. Log-Log

The linear OLS model breaks down at $n = 236$ due to high skewness, while the log-log specification achieves $R^2 = 0.497$. The original $n = 46$ linear result is replicated by the pipeline ($R^2 = 0.346$ vs. the paper's 0.361), confirming structural validity before the model transition.

C. Primary Log-Log OLS Results

TABLE III
LOG-LOG OLS REGRESSION RESULTS: PREDICTORS OF $\ln(\text{TLS}_{\text{SCALED}})$

Variable	Coeff.	SE	95% CI	p
Intercept	4.4151	0.795	[2.849, 5.981]	<0.001
$\ln(\text{TechSpend})^a$	0.4776	0.047	[0.384, 0.571]	<0.001
$\ln(\text{Employees})$	-0.3594	0.024	[-0.408, -0.311]	<0.001

$R^2 = 0.497$; Adj. $R^2 = 0.493$; $F(2, 233) = 115.04$, $p < 0.001$, $n = 236$.
^aStructural coefficient only; TechSpend is embedded in TLS by construction.

The log-log model explains 49.7% of cross-sectional variance in $\ln(\text{TLS})$ ($R^2 = 0.497$, Adj. $R^2 = 0.493$; $F(2, 233) = 115.04$, $p < 0.001$). The coefficient for $\ln(\text{TechSpend})$ is positive and significant ($\beta = 0.4776$, $\text{SE} = 0.047$, $p < 0.001$), reflecting the structural property of the index; this near-unit elasticity indicates that a 1% increase in annual technology spend is associated with approximately a 0.48% increase in TLS, consistent with construction. The coefficient for $\ln(\text{Employees})$ is negative and significant ($\beta = -0.3594$, $\text{SE} = 0.024$, 95% CI [-0.408, -0.311], $p < 0.001$), indicating that a 1% increase in employee count is associated with a 0.36% decrease in TLS, holding technology spending constant.

D. Robustness Check: Employees-Only Model

TABLE IV
EMPLOYEES-ONLY ROBUSTNESS CHECK: $\ln(\text{TLS}) \sim \ln(\text{EMPLOYEES})$,
 $n = 236$

Variable	Coeff.	SE	95% CI	p
Intercept	12.2025	0.224	[11.761, 12.644]	<0.001
$\ln(\text{Employees})$	-0.2469	0.026	[-0.298, -0.196]	<0.001

$R^2 = 0.277$; Adj. $R^2 = 0.274$; $F(1, 234) = 89.83$, $p < 0.001$, $n = 236$.

The employees-only OLS regression, the primary structural validation test carrying no mechanical relationship with TLS

by construction, confirms a significant negative association between workforce size and TLS ($\beta = -0.2469$, $\text{SE} = 0.026$, 95% CI [-0.298, -0.196], $p < 0.001$, $R^2 = 0.277$, $F(1, 234) = 89.83$, $p < 0.001$). This result is stronger than the original paper's robustness check ($\beta = -0.0731$, $p < 0.001$, $n = 46$, linear scale), and confirms that larger firms achieve lower technology leverage per labor hour regardless of whether technology spending is controlled.

V. DISCUSSION

The regression results, combined with the employees-only robustness check and sector-level construct validity evidence, provide strong structural support for TLS as a measure of technology capital intensity. Three substantive findings emerge.

A. Interpreting Your Own TLS Score

A central motivation for this framework is to give practitioners a simple reference point. Once a firm computes its own TLS, the natural question is: *is this number good?* The sector benchmarks derived from the 236-firm dataset provide a direct answer. Using the formula $\text{TLS} = 1,000 \times \text{TechSpend}/\text{AnnualLaborHours}$, the median TLS across all firms in this study is approximately **9,585**, with the 25th percentile at 4,941 and the 75th at 15,409.

Sector-specific benchmarks from this dataset are as follows. Enterprise Tech firms have a median TLS of roughly 10,554 (P25: 7,387; P75: 16,718). Cybersecurity firms cluster around a median of 9,757 (P25: 6,719; P75: 13,495). FinTech firms sit at a median of 14,793 (P25: 8,434; P75: 23,234). Consumer Tech firms have a median of 14,985 (P25: 10,675; P75: 27,091). The broad Tech category, which includes semiconductor manufacturers, hardware distributors, and IT services firms, has a lower median of 5,656 (P25: 2,734; P75: 12,363), reflecting the labor-intensive manufacturing and services operations in that group. Clean Tech sits at a median of 5,704 (P25: 4,230; P75: 13,799), consistent with capital-intensive physical infrastructure.

To use these benchmarks: compute your firm's TLS using disclosed or estimated annual technology expenditure and total annual labor hours ($\text{Employees} \times 2,080$). Then locate your sector row. A score below the P25 for your sector suggests your technology investment is low relative to headcount, which may signal under-investment or a workforce that is growing faster than digital tooling. A score above the P75 suggests high technology intensity, typical of lean, software-heavy operations or firms that have automated heavily. A score near the median indicates you are investing roughly in line with peers in your industry.

TLS should be read as a ratio to watch over time rather than a single-point judgment. A firm whose TLS rises from 8,000 to 12,000 over two years has increased its technology intensity per labor hour by 50%, regardless of whether it grew headcount or trimmed it. Conversely, a firm that doubled its technology budget but also doubled headcount will see flat TLS, meaning the investment did not improve leverage. The practical rule of thumb is: if TLS is rising, technology

is compounding faster than the workforce; if it is falling, the workforce is growing faster than technology adoption. Comparing your firm’s trajectory against the sector medians above gives an immediate peer benchmark without needing access to a proprietary database.

First, the consistent negative relationship between organizational scale and TLS, confirmed across the robustness check ($\beta = -0.247$, $p < 0.001$), the joint log-log model ($\beta = -0.359$, $p < 0.001$), and the original linear model ($\beta = -0.073$, $p < 0.001$), is the most important finding. Firms with more employees consistently exhibit lower technology leverage per labor hour, using the same level of technology expenditure. This can be attributed to coordination costs: larger organizations face more complex approval workflows, greater role specialization, and slower technology adoption cycles that dilute the per-employee productivity benefit of a given technology investment [4], [8]. Smaller firms, by contrast, often have the organizational flexibility to adopt and integrate new tools rapidly, achieving high capital intensity per labor unit without the overhead that accompanies scale. These findings are consistent with the IT productivity paradox literature [7].

Second, the ratio of technology spending to labor input, not absolute expenditure, drives leverage. Firms that grow their workforce faster than their technology budget will experience declining TLS over time. Maintaining or increasing technology investment on a per-employee basis appears to be a necessary condition for sustaining operational leverage as firms scale.

Third, the sector benchmarking results provide the clearest evidence of construct validity published for this index to date. The fact that the pipeline produces the theoretically expected sector ordering without manual calibration (Enterprise Tech and Cybersecurity highest, Clean Tech lowest) demonstrates that TLS captures real economic variation in technology capital intensity across industries.

It should be recognized that TLS is a measure of technological resources deployed per labor hour, not a direct measure of output or profitability. High TLS indicates that a firm uses a large amount of technology per labor hour; it does not indicate whether those resources were selected or utilized efficiently. Future work relating TLS to output or margin data will further validate its use as a productivity proxy.

VI. LIMITATIONS

Several limitations should be considered when interpreting these findings.

Dataset composition. The expanded $n = 236$ dataset combines 44 original collected firms with 192 newly researched real companies sourced from public SEC filings (10-K, 10-Q, 8-K, 20-F), investor relations disclosures, and financial data aggregators. While every firm in the expanded set is a real publicly traded or recently private company, the sourcing methodology for the 192 additions relies on disclosed annual figures and, for some quarterly series, modeled growth curves anchored to real disclosed endpoints. Users should treat the expanded results as robust cross-sectional benchmarks while

recognizing that data quality varies across the 236 firms; rows with `data_source = 'real_verified'` carry higher data fidelity than the original 44.

Cross-sectional design. The data are cross-sectional, precluding causal inference. The negative relationship between employee count and TLS could reflect reverse causality or omitted-variable bias from unobserved characteristics such as management quality or workforce skill. Longitudinal panel data would allow fixed-effects estimation.

Mechanical correlation. TechSpend is embedded in the TLS numerator, so the positive regression coefficient for $\ln(\text{TechSpend})$ is structurally expected and should not be interpreted as independent predictive evidence. This paper addresses this by treating the TechSpend result as structural validation only and foregrounding the employees-only robustness check as the primary empirical test.

Employees-only null finding at balanced scale. Unlike the original pilot’s significant univariate employee effect ($\beta = -0.073$, $p < 0.001$, $n = 46$), an $n = 100$ balanced test produced a non-significant univariate result ($\beta = -0.061$, $p = 0.266$). This indicates that in a larger, more balanced sample, employee count alone is an insufficient predictor of TLS. The relationship is correctly identified only in the joint model, where tech spending is controlled. This is an informative methodological finding, not a failure, but it qualifies the original paper’s robustness claim.

Sector homogeneity. The original 44 collected firms are drawn from four technology sub-sectors, which limits generalizability to manufacturing, retail, healthcare, and other industries. Cross-industry validation is a priority for future work.

Self-reported technology use. Technology expenditure and usage data were self-reported, introducing potential measurement error and social desirability bias. More objective measurement approaches, such as system log analysis or IT asset management records, would improve construct validity.

VII. CONCLUSION

This paper introduces the Technology Leverage Score (TLS), a quantitative index that measures technology capital intensity per labor unit using standard firm-level operational data. Applied to an expanded dataset of 236 technology-sector firms spanning semiconductor, enterprise software, cybersecurity, FinTech, clean energy, and consumer technology companies, TLS exhibits substantial cross-sectional variation and is significantly associated with firm size in the theoretically predicted direction across multiple specifications: a log-log OLS model ($\beta_{\ln(\text{Employees})} = -0.359$, $p < 0.001$, $R^2 = 0.497$) and an employees-only robustness check free of mechanical confounding ($\beta = -0.247$, $p < 0.001$). Sector benchmarking confirms construct validity: Enterprise Tech and Cybersecurity achieve the highest median TLS values, Clean Tech achieves the lowest, exactly as predicted by economic theory.

The single largest implication is that technology capital intensity, the ratio of technology to labor, has a greater impact on operational leverage than the absolute levels of either resource.

Companies whose workforce grows faster than technology adoption will experience declining TLS over time. To maintain or improve technology leverage as a company grows, it must purposefully manage its technology-to-employee ratio rather than simply increase spending.

The TLS framework provides a reproducible, data-driven measurement method implementable by SMBs without proprietary benchmarking tools or surveys. The analytical pipeline can be run directly by companies: provide firm-level data in the template format, execute the pipeline, and receive TLS scores, regression benchmarks, and sector comparisons. Future work should examine the long-term scale-leverage relationship using longitudinal panel data, extend validation to non-technology industries, and develop multidimensional TLS variants that separately capture returns by technology category (infrastructure, analytics, automation, communication).

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